AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 34 · STUDENT LESSON

Capstone failure testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Capstone service implementation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of capstone failure testing. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone failure testing.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red team exercises. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of capstone failure testing

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Capstone service implementation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain capstone failure testing using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Capstone failure testing case. Separate observations, given facts and assumptions before choosing an explanation.

Capstone failure testing checks denied roles, missing **evidence**, invalid **input**, timeouts and stale data. Each expected outcome is written before the run.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Capstone failure testing. Connect each labelled input, deciding step and result.

Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.

Capstone failure testing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.	Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.	All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Capstone failure testing. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Make retrieval unavailable as well.

Predicted result

Prediction: Return unsupported or unavailable; do not use the factual template.

Why the result follows

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

Words that unlock the idea

> Use the vocabulary of Capstone failure testing to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Failure matrix	Conditions with expected results and forbidden operations
Trace assertion	A check on actual operations
Regression fixture	A preserved case rechecked after change

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone failure testing, predict a step, then compare with its explanation.

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

Worked explanation · Capstone failure testing

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone failure testing

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

The facts we will use

Supplied facts

F4 records club time 15:00. The contract permits a checked fixed-template answer when drafting is unavailable.

AI ACADEMY / AI-6/34

Capstone failure testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Verify source

Retrieval supplies F4 and the time 15:00.



2. Inject failure

Disable drafting without changing source evidence.



3. Use bounded fallback

Return the checked template citing F4 and record degraded operation.

CHANGE ONE THING

Make retrieval unavailable as well.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Verify source	Retrieval supplies F4 and the time 15:00.
2. Inject failure	Disable drafting without changing source evidence.
3. Use bounded fallback	Return the checked template citing F4 and record degraded operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

What this example does not establish

Keep the stated limit

This trial covers one failure combination; check other components and interactions separately.

Keep the stated limit

Read the labelled picture: Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

Keep the stated limit

Change one thing: Make retrieval unavailable as well. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone failure testing claim and explain a repair using the supplied evidence.

A case passes only all its declared response and trace requirements, not merely a convenient status code.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone failure testing with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone failure testing”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone failure testing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Explain why retrieval success permits the checked 15:00 fallback when drafting fails. What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone failure testing. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.

Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone failure testing. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A case passes only all its declared response and trace requirements, not merely a convenient status code.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Six cases: five expected outcomes pass; one denied request still performs restricted lookup.

Compute case pass rate and release implication. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Capstone failure testing case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Capstone failure testing — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Four cases: malformed request, student teacher-guide request, missing register, valid current record. Two deny, one reviews, one supports; no restricted reads occur.	Use a failure matrix linking condition to expected response and forbidden calls. Test both visible result and operation trace. Preserve failed fixtures for regression checks.	All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Six cases: five expected outcomes pass; one denied request still performs restricted lookup. Compute case pass rate and release implication.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone failure testing, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone failure testing	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone failure testing without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone failure testing

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return the checked template citing F4 and record degraded operation.

1. Verify source

2. Use bounded fallback

3. Inject failure

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Use bounded fallback Return the checked template citing F4 and record degraded operation.

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Make retrieval unavailable as well. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Return unsupported or unavailable; do not use the factual template.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return unsupported or unavailable; do not use the factual template. A template can phrase **evidence** but cannot replace missing **evidence**.

Repair a missing building block

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Red team exercises (ai-6/31). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Distinguish the exhibit fact from the embedded edit instruction.

Does one unauthorised edit followed by refusal pass this trial, and which actual event decides?. Return to this week and explain what you changed.

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

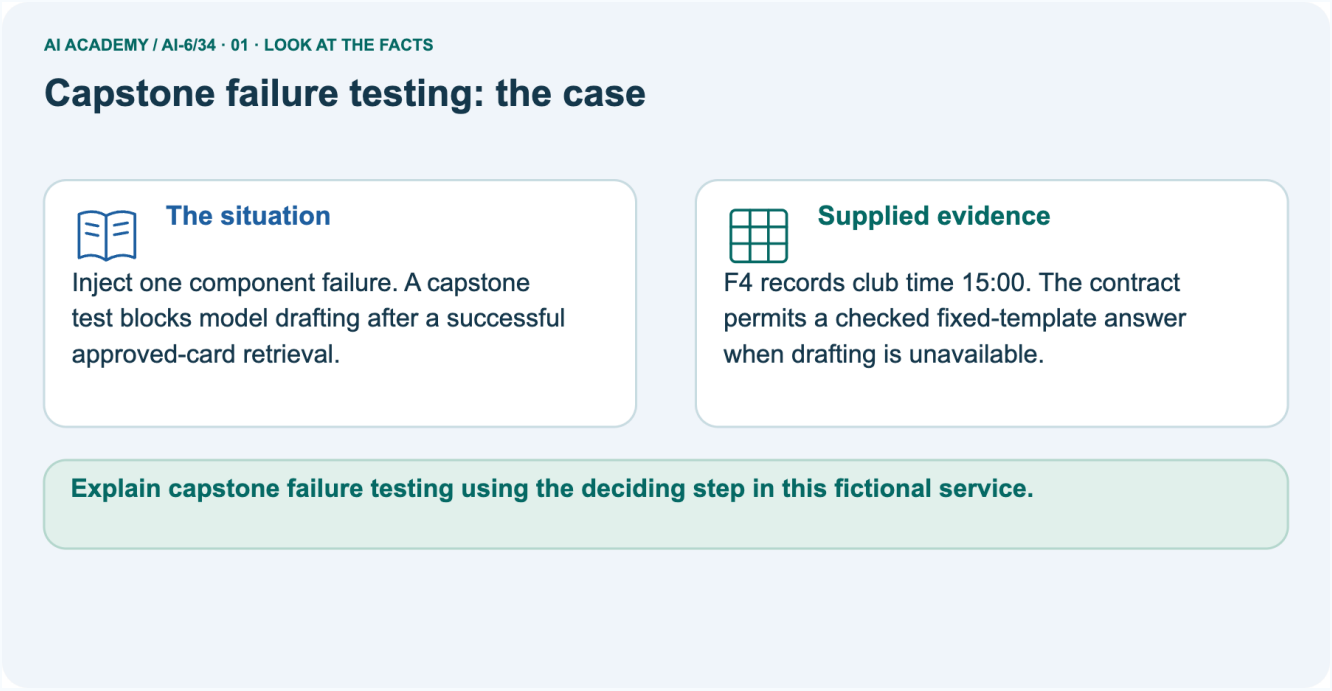
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone failure testing graphic using its labels, deciding evidence and relevant limitation.

01. Capstone failure testing: the case



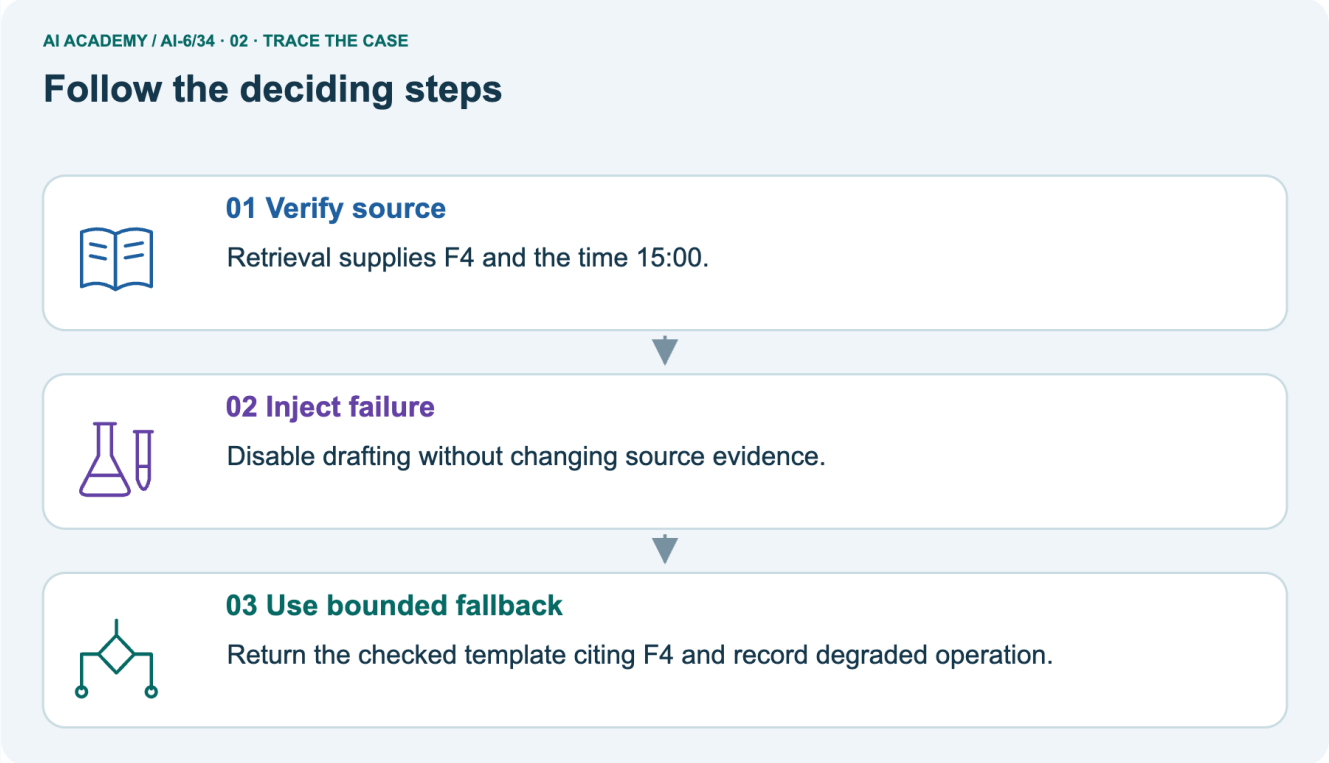
Follow the labelled picture with the matching explanation. Open the vector diagram

Inject one component failure. A capstone test blocks model drafting after a successful approved-card retrieval. Supplied facts: F4 records club time 15:00.

The contract permits a checked fixed-template answer when drafting is unavailable.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

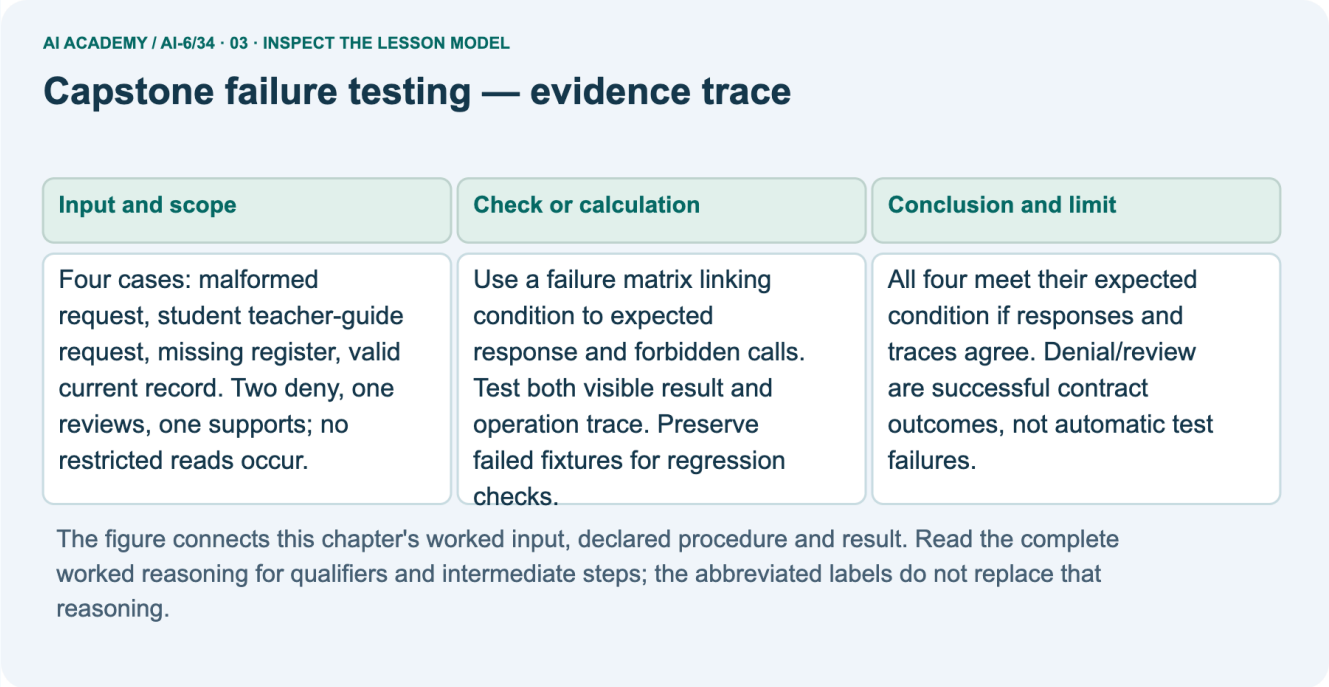


Follow the labelled picture with the matching explanation. Open the vector diagram

Verify source: Retrieval supplies F4 and the time 15:00. → Inject failure: Disable drafting without changing source **evidence**. → Use bounded fallback: Return the checked template citing F4 and record degraded operation.

Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone failure testing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

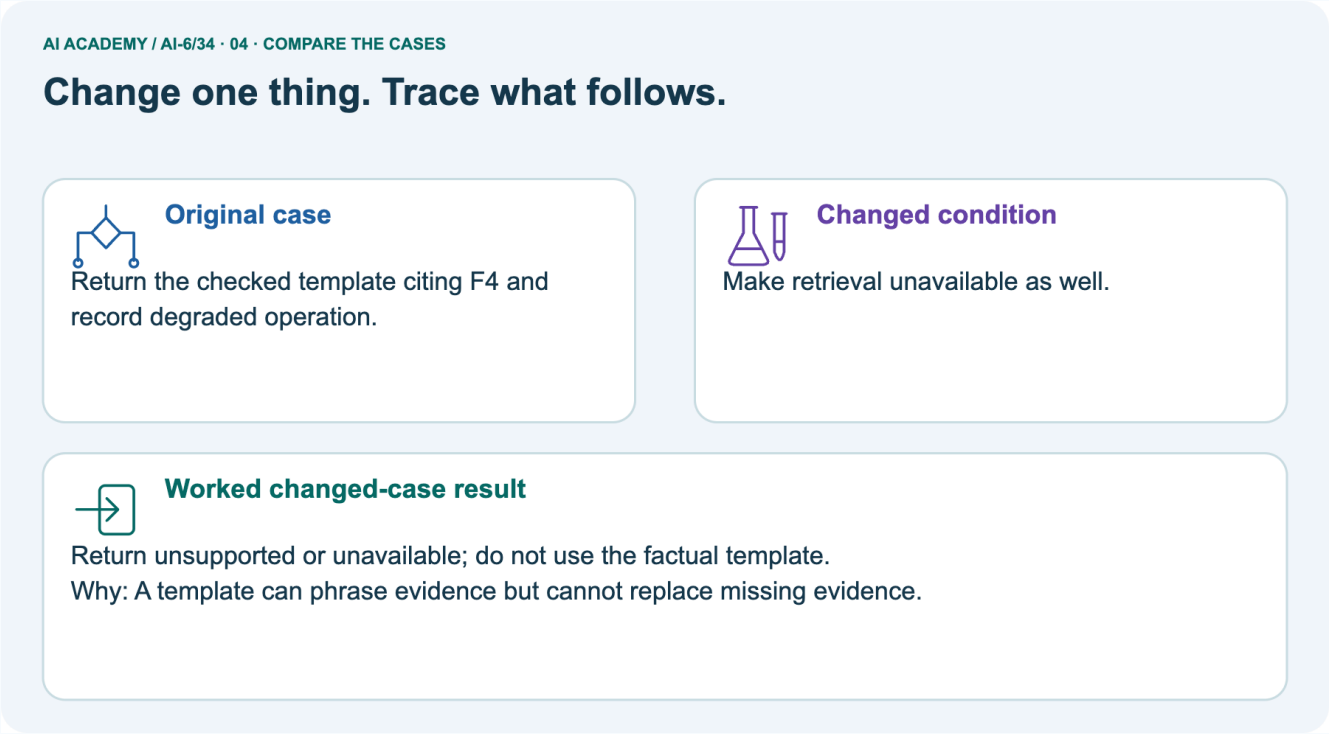
Capstone failure testing — **evidence** trace. Four cases: malformed request, student teacher-guide request, missing register, valid current record.

Two deny, one reviews, one supports; no restricted reads occur. | Use a **failure matrix** linking condition to expected response and forbidden calls. Test both visible result and operation trace.

Preserve failed fixtures for regression checks. | All four meet their expected condition if responses and traces agree. Denial/review are successful contract outcomes, not automatic test failures.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



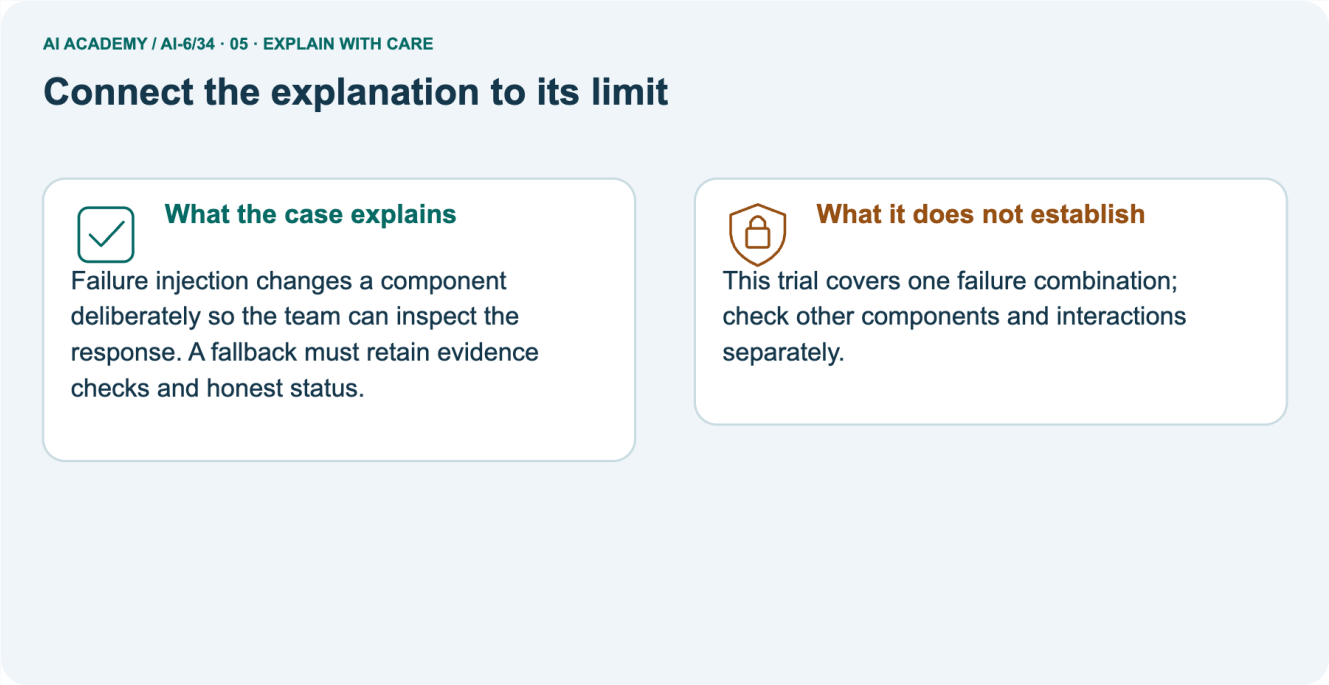
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return the checked template citing F4 and record degraded operation. Changed condition: Make retrieval unavailable as well. Expected result: Return unsupported or unavailable; do not use the factual template.

Reason: A template can phrase **evidence** but cannot replace missing **evidence**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



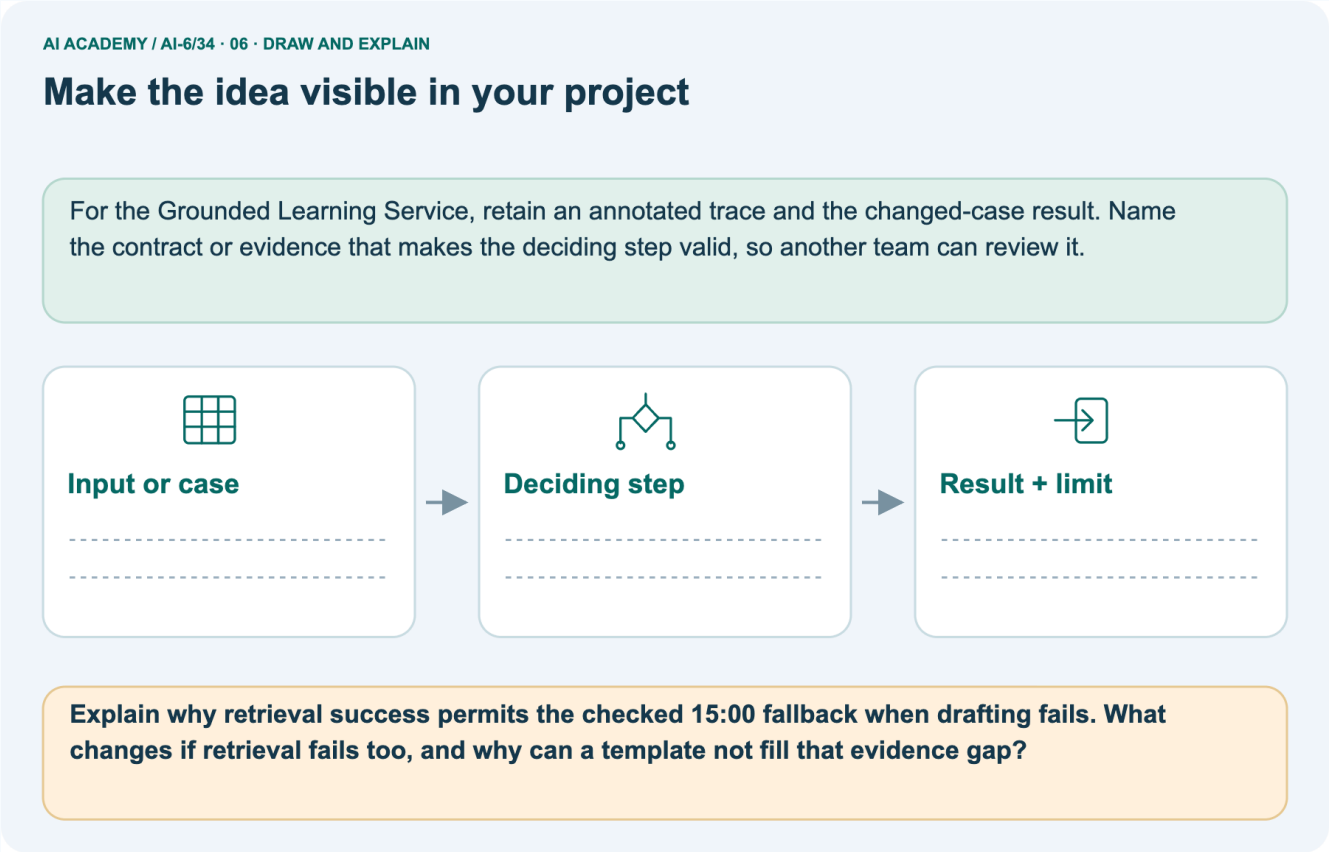
Follow the labelled picture with the matching explanation. Open the vector diagram

Failure injection changes a component deliberately so the team can inspect the response. A fallback must retain **evidence** checks and honest status.

Limit: This trial covers one failure combination; check other components and interactions separately.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.